

MINERAL ECONOMICS AND POLICY

John E. Tilton and Juan Ignacio Guzmán



Mineral Economics and Policy

This book provides an introduction to the field of mineral economics and its use in understanding the behavior of mineral commodity markets and in assessing both public and corporate policies in this important economic sector. The focus is on metal and non-metallic commodities rather than oil, coal, and other energy commodities.

The work draws on John Tilton's teaching experience over the last 30 years at the Colorado School of Mines and the Catholic University of Chile, as well as short courses for Rio Tinto and other mining companies. This is combined with the professional consulting and academic research of Juan Ignacio Guzmán over the past decade, in order to demonstrate the industry application of the economic principles described in the earlier chapters.

The book should be an ideal text for graduate and undergraduate students in the fields of mining engineering and natural resource economics and policy. It should also be of interest to professionals and investors in mining and commodity markets, and those undertaking continuing education in the mineral sector.

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John E. Tilton and Juan Ignacio Guzmán

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Preface

The origins of this book date back to the fall of 1972, when I joined the Department of Mineral Economics at the Pennsylvania State University. A decade or so earlier I had written my PhD dissertation on the determinants of international trade patterns for a number of major nonferrous metals. But only at Penn State did I start teaching a course on metal markets and industries, and so thinking about the metal industries and the field of mineral economics in a comprehensive manner. How do the supply and demand for mineral commodities differ between the short run and the long run? And, how do they differ from the supply and demand for agricultural or manufactured products? What can one say about the behavior of mineral prices and markets—why are they so volatile, for example? What important issues for public policy arise from the mineral sector, and what insights, if any, can simple economic tools shed on these issues?

Back then, governments worried about strategic and critical materials, particularly those such as cobalt and manganese largely coming from southern Africa and the Soviet Union. Depletion and the long-run availability of mineral commodities was another widespread concern, particularly in light of their non-renewable nature and the expected growth in the global economy. Even before OPEC (the Organization of Petroleum Exporting Countries) pushed up oil prices in 1973, cartels, monopolies, and mergers and acquisitions attracted considerable attention. While sustainable development was not an issue, certainly the pollution and environmental damage associated with mining and mineral production were.

Since 1972 I have continued to teach my course on mineral economics with its focus on metal and other non-energy mineral commodities. Over time the course has evolved as my teaching shifted from Penn State to the Colorado School of Mines and more recently to the Catholic University of Chile. The changes reflect in part the waxing and waning of the policy issues clamoring for public attention associated with mining and the mineral sector. They also reflect my own understanding of the mineral industries and the important economic, institutional, and technological forces governing their behavior.

Few academics have the opportunity to develop and teach over several decades their own specialized course. To my knowledge my course is unique with no close substitutes. For this reason, a decade or so ago, about the time I officially retired from the Colorado School of Mines, I began thinking about writing this book.

When I really retired, the book would provide those with an interest in mineral economics access to the conceptual framework and other material that I have found useful for understanding mineral commodity markets.

As the years slipped by with only a few chapters emerging into daylight, I began to realize I needed help from a good disciplinarian. It was then that I enlisted Juan Ignacio Guzmán to join me in this endeavor. I met Juan Ignacio while serving on his PhD dissertation committee at the Catholic University of Chile in 2007 and was impressed with both his command over the modern tools of economics and his in-depth understanding of copper and other commodity markets. Upon completing his dissertation, Juan Ignacio worked for several years in the Santiago office of CRU, a well-known consulting group focusing on mineral commodity markets. He then started his own, very successful consulting company, GEM (*Gestión y Economía Ltda.*). While working full-time, he has also continued to teach part-time at several universities in Santiago, including as a colleague at the Catholic University of Chile.

Juan Ignacio is the ideal partner. He complements my background with his strong knowledge of recent theoretical and quantitative developments in the field of economics, his interest in resource economics and industrial organization, and his understanding of mineral commodity markets based on extensive experience as an industry consultant. While my interests largely focus on appropriate public or government policies for the mineral sector, Juan Ignacio brings a strong interest in private or firm policies as well. Finally, let me also confess that, should this book find a useful niche helping interested readers better understand the fascinating machinations of metal and mineral commodity markets, I am hoping Juan Ignacio will from time to time provide updated and revised editions of this volume, as the importance of the mineral sector for welfare of humanity is not likely to wane over time.

Let me finish this preface by acknowledging and thanking some of those who helped make this volume possible. A complete list here is impossible, since we both have benefited over the years from the knowledge and insights of the many students we have taught and advised, of our colleagues, and of the engineers and other professionals with whom we have had the opportunity to interact. There are a few names, however, that we must mention.

First, we want to thank Tim Hardwick, our Senior Editor at RFF Press, for his enthusiasm and gentle prodding over the lengthy gestation period of this project, and Ashley Wright, Hannah Champney, Dave Wright, and Gary Smith for their assistance in the subsequent editorial and production processes. In addition, we are grateful to Robert D. Cairns, Phillip Crowson, Graham A. Davis, Roderick G. Eggert, Joaquín Jara, David Humphreys, Gustavo Lagos, Philip Maxwell, and Marian Radetzki for their useful insights and helpful comments on earlier drafts.

John E. Tilton
Golden, Colorado
September 30, 2015

1 Introduction

The men and women who extract mineral resources from the earth's crust and process them into a variety of different metals and other mineral commodities make life as we know it possible. It has not always been so. For millennia our distant ancestors lived off the land, the surface of the land, first from hunting and gathering, and then from raising their own crops and animals. Only about 40,000 years ago did this change when humans began on a significant scale extracting flint and useful minerals from within the earth. The subsequent division of history into the Stone Age, Bronze Age, and Iron Age highlights the importance of mining from that time onward.

Moreover, over the past several centuries the impact of mineral commodities on society has exploded. This period has witnessed the discovery and commercialization of many new materials, including aluminum, steel, chromium, lithium, and the rare earth minerals. Even taking into account the materials widely used as far back as the Roman era—copper, tin, lead, stones, and clays, for example—it is hard to think of a single mineral commodity whose consumption over the past 50 years does not exceed by a wide margin the quantities used throughout all previous history. These resources from the earth are needed to build homes, roads, bridges, offices, schools, and hospitals. They are found in computers, food and beverage containers, automobiles, airplanes, modern windmills, and the lowly 50-watt light bulb. Without them, modern industry, agriculture, communications, transportation, science, and medicine would simply not exist.

Mineral Economics and Policy

This book—*Mineral Economics and Policy*—provides an overview of how mineral markets and industries operate and why they behave as they do. The two of us together have devoted more than half a century to teaching, researching, and consulting on economic and policy issues associated with mining and with the production and use of mineral commodities. *Mineral Economics and Policy* draws on this work, synthesizing, updating, distilling, and extending it. It is written for both the specialist and the interested layperson and requires only an introductory understanding of supply, demand, and other basic economic principles. While far

2 Introduction

from comprehensive in the sense of treating all interesting economic and policy issues associated with the mineral industries and markets, the book does provide a conceptual framework that many of our students and we have found helpful. It then applies this framework to examine some of the interesting policy issues arising from the mining, processing, and use of mineral commodities.

The focus is on the metals and to a somewhat lesser extent the non-metallics. Among the metals some will detect a bias in favor of copper. This simply reflects our own interests over the years and the fact that both of us teach in the Mining Engineering Department at the Catholic University of Chile. Copper is tremendously important in Chile, accounting in recent years for nearly half of the country's export earnings and a large share of its government revenues as well. Nevertheless, much of the analysis applies as well to mineral commodities more generally, including diamonds, phosphate rock, and even petroleum. In this respect, *Mineral Economics and Policy* provides an introduction to the field of mineral economics.

Defined broadly, as is usually the case in North America, mineral economics covers three classes of commodities—energy, non-metallics, and metals. The energy minerals include coal, petroleum, natural gas, and uranium, and are by far the most important of the three groups in terms of sales. They are also the most studied. Many good economic analyses of energy markets exist, including several textbooks on energy economics.¹ By comparison, the metals and the non-metallic mineral commodities have received far less attention.²

Among the metals, iron and steel, aluminum, copper, gold, nickel, lead, zinc, and tin are the most important, at least in terms of sales (see [Table 1.1](#)). Many others also exist, whose importance should not be underestimated. Modern electronics, for example, is hard to imagine without semiconductors made of silicon or without liquid crystal displays and other flat-panel devices made with indium.

The non-metallics include construction materials (such as sand and gravel, stone and marble), fertilizers (such as potash and phosphate rock), and all the other mineral commodities that are neither metals nor energy resources (see [Table 1.1](#)). Their production when measured by volume or tonnage exceeds that of the metals. Nevertheless, the non-metallics have received even less attention from policy analysts and others than the metals. Just why this is so is not entirely clear. Part of the explanation may lie in the fact that they are for the most part large-volume, low-cost products often sold in regional or local markets. They enter international trade on a much smaller scale, and typically require less sophisticated technologies to extract and process. This, though, is far from a satisfactory explanation, and in any case numerous non-metallic products do not fit this simple generalization, as phosphate rock and diamonds so amply illustrate.

The mineral industries, it is important to highlight, produce a vast array of quite different substances and products. Some mineral commodities are extracted from large, open pits; others are hoisted out of underground mines; still others are pumped from deep wells; and a few are processed from seawater. Production can be simple and cheap, though often it is technically complicated and costly. Some mineral commodities are recovered as by-products or co-products in the course of

extracting other commodities. Some are obtained from recycling old and new scrap. Some are found in only a few locations and are traded worldwide. Others are produced in many locations and consumed close to their terrestrial source. Some are traded on competitive exchanges where prices fluctuate daily. Others are produced by only a few firms and are offered at stable producer prices.

Mineral Economics and Policy does examine non-metallic mineral commodities, though it too is to some extent guilty of paying insufficient attention to the non-metallics, perpetuating an injustice to an important group of mineral commodities. The reason is simply that our own interests, as noted, have over the years largely focused on the metals. Fortunately, much of the conceptual framework and many of the policy issues explored in *Mineral Economics and Policy* are just as relevant and useful for the non-metallics as the metals.

This rich diversity makes mineral commodities and the field of mineral economics fascinating to study, but it also means no general model or economic analysis is applicable to all mineral commodities or even all the metals. Rather, each must be considered individually, thereby allowing the analysis to take account of its particular features.

As a result, a single book cannot begin to cover all the metals, let alone all mineral commodities, and no attempt is made to do so here. Instead, *Mineral Economics and Policy* concentrates on illustrating the usefulness of relatively simple economic principles, particularly those associated with supply and demand, in understanding the behavior of commodity markets.

A good mineral economist must possess a strong understanding of the technological and institutional conditions that shape and constrain the behavior of mineral commodity markets. What are the stages in the production of steel? How does mining law vary around the world and how, in turn, does it affect exploration? How capital intensive is the production of aluminum? How are the creation and diffusion of new technologies, such as the solvent extraction electrowinning process, altering the production and price of copper and other mineral products? How are they changing the associated environmental pollution?

For this reason, we believe mineral economics is more than just a specialty within economics. It is an interdisciplinary field that draws as well on geology, mining engineering, environmental science, metallurgy, and other associated technical fields.

There is one particularly important conclusion that the pages that follow highlight time and time again. The simple tools of economics can provide powerful insights into the nature and behavior of the mineral markets, but only if the analyst applying these tools has a firm understanding of the important technological and institutional relationships governing the market he or she is examining, and so can tailor the analysis to take these relationships explicitly into account.

The Road Ahead

Mineral Economics and Policy is divided into two parts. [Part I](#) develops the conceptual framework. Specifically, it assesses mineral commodity demand,

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supply, markets, and prices. In the process, it explores the nature of material substitution, recycling, and secondary production, along with by-product and co-product output. It analyzes the important forces shaping demand in both the past and the future. It examines the relationships between production costs, technological change, and supply. It describes how mineral commodity markets have evolved over the past half century and investigates the volatility of their prices in the short run and their secular trends over the long run.

Part II explores policy issues associated with mining, mineral processing, and use that are of particular interest to governments and other entities concerned with the welfare of society in general. In particular, it considers:

- How mining and metal production produces economic rents and the role of taxation and other policies in the distribution of this wealth.
- The reasons why certain countries enjoy a comparative advantage or competitiveness in the production and export of iron ore or other mineral commodities, while other countries rely on imports for all or most of their needs.
- The nature of market power and the evolution of antitrust or competitiveness policies in the mineral sector over time.
- The troubling relationship between mineral wealth and economic development, including the nature of the Dutch Disease and the arguments both for and against the Resource Curse.
- The long-run threat from mineral depletion and the important differences between the major mental models used to assess this threat—the fixed-stock paradigm and the opportunity-cost paradigm.
- Environmental and sustainability issues—such as intergenerational equity and the socially optimal amount of recycling, conservation, and substitution of renewable for non-renewable resources—associated with mineral production and use.

Notes

- 1 For example, see Dahl (2014).
- 2 There are fortunately a few good exceptions. See Crowson (2008), Humphreys (2015), Maxwell (2013), and Radetzki (2008). Though these, we believe, are good complements rather than substitutes for this volume.

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Table 1.1 Classification, end products, and 2014 sales in millions of US dollars for selected mineral commodities

Classification		Commodity	Products	2014 sales [million \$]
Metals	Ferrous metals*	Iron ore	Concentrates	325,220
			Coarse ores	
			Fines ores	
			Pellets	
	Nonferrous metals	Aluminum	Briquettes	
			Sinter	
			Roasted iron pyrites	
			Unwrought (in coils)	120,238
			Unwrought (other than aluminum alloys)	
			Unwrought (billet)	
			Bauxite, calcined	
	Copper	Copper	Bauxite, crude dry (metallurgical grade)	129,862
			Alumina	
			Aluminum hydroxide	
			Waste and scrap	
	Tin	Tin	Copper ores and concentrates	6,395
			Unrefined copper anode	
			Refined and alloys (unwrought)	
			Copper wire (rod)	
	Nickel	Nickel	Unwrought tin, not alloyed	40,471
			Unwrought tin, containing lead	
			Tin waste and scrap	
			Nickel oxides, chemical grade	
			Ferronickel	
			Unwrought nickel, not alloyed	

Table 1.1 *continued*

<i>Classification</i>		<i>Commodity</i>	<i>Products</i>	<i>2014 sales [million \$]</i>
Metals	Nonferrous metals	Base	Unwrought (refined) Antimonial lead Alloys of lead	10,352
			Zinc ores and concentrates Zinc oxide and zinc peroxide Unwrought zinc, not alloyed Zinc alloys	28,881
	Light	Beryllium	Beryllium ores and concentrates Beryllium oxide and hydroxide Beryllium-copper master alloy Beryllium unwrought, including powders Waste and scrap	121
			Lithium oxide and hydroxide Lithium carbonate Concentrates and others	1,265
			Titanium oxides (unfinished TiO ₂ pigments) TiO ₂ pigments (80 percent or more TiO ₂) Ferrotitanium and ferrosilicon titanium Unwrought titanium metal Other titanium metal articles Wrought titanium metal Synthetic rutile Ilmenite and ilmenite sand Rutile concentrate Titanium slag Titanium waste and scrap	4,982

<i>Classification</i>		<i>Commodity</i>	<i>Products</i>	<i>2014 sales [million \$]</i>
Metals	Nonferrous metals	Platinum group metal (PGM)	Iridium	582
			Iridium unwrought, including powders	
		Palladium	Unwrought and semi-manufactured	5,070
	Precious	Gold	Unwrought gold, including bullion and doré	116,778
		Platinum	Unwrought and semi-manufactured	7,454
		Silver	Unrefined silver Refined bullion	15,969
	Refractory	Cobalt	Cobalt ores and concentrates	3,457
			Chemical compound:	
			Cobalt oxides and hydroxides	
			Cobalt chlorides	
			Cobalt sulfates	
			Cobalt carbonates	
			Cobalt acetates	
			Unwrought cobalt alloys	
			Cobalt mattes and other intermediate products, cobalt powders	
			Wrought cobalt and cobalt articles	
			Waste and scrap	

Table 1.1 *continued*

<i>Classification</i>	<i>Commodity</i>		<i>Products</i>	<i>2014 sales [million \$]</i>
Metals	Nonferrous metals	Refractory	Molybdenum	7,155
			Molybdenum ore and concentrates	
			Molybdenum chemicals	
			Molybdenum oxides and hydroxides	
			Molybdates of ammonium	
			Molybdates, all alloys	
			Molybdenum pigments	
			Ferromolybdenum	
			Molybdenum metals	
			Powder	
			Unwrought	
			Wrought planes, sheets, trips, etc.	
			Wire	
			Waste and scrap	
	Niobium		Synthetic tantalum–niobium concentrates	3,812
			Niobium ores and concentrates	
			Niobium oxides	
			Ferroniobium	
			Niobium unwrought, alloys, metal, powders	
			Waste and scrap	
	Rhenium		Salts of peroxometallic acids	146
			Rhenium unwrought, powders	
			Rhenium wrought	
			Waste and scrap	
	Tantalum		Synthetic tantalum–niobium concentrates	355
			Tantalum ores and concentrates	
			Tantalum oxides	
			Unwrought: powders and metal alloys	
			Waste and scrap	

<i>Classification</i>	<i>Commodity</i>		<i>Products</i>	<i>2014 sales [million \$]</i>
Metals	Nonferrous metals	Refractory	Tungsten	3,637
			Tungsten ores and concentrates	
			Tungsten oxides	
			Ammonium tungstates	
			Tungsten carbides	
			Ferrotungsten	
			Tungsten powders	
			Zirconium	
			Zirconium ores and concentrates	1,494
			Germanium oxides and zirconium dioxide	
			Ferrozirconium	
			Unwrought and zirconium powders	
			Other zirconium articles	
			Waste and scrap	
	Nonclassified	Arsenic	Metal	30
			Acid	
			Trioxide	
			Sulfide	
			Bismuth	213
			Bismuth and articles thereof	
			Waste and scrap	
			Cadmium	43
			Cadmium oxide	
			Cadmium sulfide	
			Pigments and preparations based on cadmium compounds	
			Unwrought cadmium and powders	
			Wrought cadmium and other articles	
			Waste and scrap	
			Chromium	6,467
			Ore and concentrates	
			Ferchromium	
			Ferchromium silicon	
			Chromium metal, unwrought, powder	
			Waste and scrap	

Table 1.1 *continued*

Classification	Commodity		Products	2014 sales [million \$]	
Metals	Nonferrous metals	Nonclassified	Indium	Unwrought indium, including powders	582
			Manganese	Ores and concentrates	36
				Manganese dioxide	
				High-carbon ferromanganeses	
				Silicomanganese	
				Metal, unwrought	
			Mercury	Mercury	100
		Silicon	Silicon	26,530	
			Ferrosilicon		
			Ferrosilicon manganese		
Non-metals	Construction	Strontium		Celestite	16
				Strontium metal	
				Strontium oxide, hydroxide, peroxide	
				Strontium nitrate	
				Strontium carbonate	
		Vanadium		Vanadium pentoxide anhydride	1,780
				Vanadium oxides and hydroxides	
				Vanadates	
				Ferrovandium	
				Vanadium and articles thereof	
	Cement		Cement clinker	411,730	
			White portland cement		
			Aluminous cement		
			Other hydraulic cement		

<i>Classification</i>	<i>Commodity</i>	<i>Products</i>	<i>2014 sales [million \$]</i>
Non-metals	Construction	Gypsum	2,214
		Salt (sodium chloride)	21,957
		Sulfur	6,878
		Crude or unrefined Sublimed or precipitated Sulfur acid	
	Fertilizers	Phosphate rock	19,800
		Potash	25,550
		Potassium nitrate Potassium chloride Potassium sulfate Potassic fertilizers Potassium–sodium nitrate mixtures	
	Refractory	Graphite (natural) Powders	1,480
	Abrasive	Industrial diamond	800
		Industrial miners' diamonds, carbonados Industrial diamonds, simply sawn, cleaved, or bruted Industrial diamonds, not worked Grit or dust and powder of natural or synthetic diamonds	
Insulate	Asbestos	Crocidolite Amosite Chrysotile crudes or milled fibers	3,089

Table 1.1 *continued*

<i>Classification</i>	<i>Commodity</i>		<i>Products</i>	<i>2014 sales [million \$]</i>
Non-metals	Pigment	Barite	Ground barite	1,158
			Crude barite	
			Oxide, hydroxide, and peroxide	
			Chlorides or sulfates or barium carbonate	
	Gemstones	Precious	Diamond, unworked or sawn	25,000
	Nonclassified	Germanium	Germanium oxides	314
			Metal, unwrought	
			Metal, powder	
		Helium	Metal, wrought	
			Helium	688

Source: GEM, based on information from the US Geological Survey.

Note:

*There is no uniform classification for the ferrous metals. Here we define the ferrous metals as those containing iron. They include iron, pig iron, cast iron, wrought iron, and steel alloys. Other classification systems consider manganese, cobalt, chromium, and other metals used in producing steel alloys as ferrous metals.

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